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How Integrated Are Rice Markets in Asia? Effects of Crises and Rice Export Quality on Price Shock Transmission

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ABSTRACT

Rising and unstable prices of rice and other staple foods threaten the food security of millions worldwide. These fluctuations raise concerns about market response to price changes, especially regarding incentives for food production. This study investigates how the price dynamics in the Asian rice markets are affected by crises in low-quality rice by analysing monthly export price data for 25% broken rice for Vietnam, Thailand, India, and Pakistan. We test the rice price transmission and rice market integration of these countries using a vector autoregression (VAR) model. The results reveal a tendency for price signals to move together across our sample countries, despite variations in their rice production and consumption patterns. This suggests that rice price shocks are transmitted between Asian exporting countries, particularly for low-quality rice. Furthermore, our analysis suggests that price dynamics in these markets are primarily influenced by crises affecting domestic agro-food chains and government policies related to price and trade. To improve production incentives during crises, governments could promote competition among traders, wholesalers, and input providers, and support farmers' income through supply oriented policies, including input vouchers and agricultural credit. These policies can mitigate price and trade distortions that can negatively affect price incentives and food security.

JEL Classification: C32, M21, Q13

1 | Introduction

Understanding how rice markets function and interact across Asia is a crucial research and policy priority. This is particularly important because rice is central to food security in many countries within the region, especially developing ones. As a staple food for billions, ensuring a stable and affordable supply of rice is vital for reducing vulnerability to food crises. However, two significant research gaps remain. First, there is limited understanding of how the rice supply and demand chain is structured and how it evolves in response to market forces. Second, the role

of rice trade and its associated value-chain dynamics in maintaining supply stability remains underexplored. These gaps underscore the need to assess the integration of rice markets to better understand the resilience and vulnerability of rice supply systems in Asia. Recent global crises have intensified volatility in the prices and availability of agricultural commodities, with significant impacts on food insecurity and nutrition (FAO 2011). Economic shocks, such as those caused by pandemics, trade restrictions, and geopolitical tensions, are now recognised to be major drivers of food crises (Ferguson and Ubilava 2022; FSIN and Global Network Against Food Crises 2023). Given that Asia

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is home to the world's largest rice exporters and importers, the price policies and trade restrictions by countries in this region strongly influence global rice market dynamics.

In 2007, India and Vietnam introduced quantitative restrictions to trade on non-basmati rice (Abbott 2012). However, Vietnam lifted this policy in 2008 (Sekhar 2012), while India has again implemented restrictions on non-basmati (non-parboiled) rice in the second half of 2023 (Valera et al. 2024). In addition, Russia and the United Arab Emirates introduced temporary export bans on 29 July 2023 and 28 July 2023, respectively. These trade restrictions could exacerbate price variations, creating a highly volatile global rice market with similar characteristics to the one experienced during the food price crisis between 2006 and 2008 (FAO 2011). While the global market features numerous rice quality grades with distinct categories, price shocks originating in one country may not impact all rice categories in other countries equally. For instance, premium aromatic rice might react differently to price changes compared to standard white rice.

This study investigates the dynamics of rice price shock transmission across selected Asian markets, focusing on the impact of crises affecting low-quality rice classes, crucial for the food security of the most vulnerable population groups. Building on these knowledge insights, we examine how different crises and geographical locations within Asia could influence the speed of price transmission for low-quality classes of rice (Valera et al. 2023). Understanding the impact of different crises and rice qualities on price transmission is critical for policymakers seeking to support farmers and consumers amidst volatile economic and international trade conditions that can exacerbate price fluctuations and food insecurity. Existing research acknowledges that price transmission in agricultural markets, including rice, is imperfect (Morales et al. 2017; Aruga and Li 2016) and exhibits asymmetrical adjustments (Hassanzoy et al. 2016). Higher degrees of price transmission imperfection can affect production incentives, reducing market efficiency and increasing price volatility in domestic markets (FAO 2011; Ferguson and Ubilava 2022; Onumah et al. 2022). While previous studies recognise the existence of diverse rice quality classes and varying consumer preferences, the impacts on price transmission among low-quality rice have not been sufficiently explored. Furthermore, we analyse the impact of major food crises, including the 2008 food price crisis, the COVID-19 pandemic and the ongoing war between Russia and Ukraine on rice price transmission. While research suggests similar price trends across major rice exporting countries during these periods (Jayasuriya et al. 2012; Akhter 2017; FSIN and the Global Network Against Food Crises 2023), we investigate potential variations in price transmission dynamics during such critical events. Understanding these nuances is crucial for informed policy decisions that address the specific impact on countries exporting and consuming different rice quality grades during periods of price volatility and food price crises.

The Asian rice market represents the heart of the global rice trade. This research focuses on this region due to its main characteristics. First, Asia dominates rice production in the world and provides the staple food for billions of people globally (FAO 2020). The region also houses both the biggest exporters

(Vietnam, India, Thailand, and Pakistan), accounting for 70% of global exports in 2020 (FAO 2020), and the greatest importers (China, Indonesia, and The Philippines), consuming 42% of global imports that same year (FAO 2020). Second, high-value demand for rice is evident in premium prices that consumers in Asian markets are willing to pay for high-quality grades (Grimm et al. 2023). However, the price transmission mechanisms for low-quality rice grades remain poorly understood. Third, price interconnectedness is key. Research confirms a “co-movement” of Asian rice prices (Chen and Saghalian 2016), meaning price changes ripple across the region, and spatial price transmission exists (John 2014).

The article proceeds as follows. In Section 2, we lay out the theoretical framework of spatial price transmission, considering how it applies to different rice grades. Section 3 examines rice price trends in key exporting countries over the past 23 years. Building on our understanding of this context, Section 4 introduces and explains our empirical testing framework for price transmission. Section 5 leverages real-world data to analyse the level of market integration among Asian rice-exporting nations. We employ econometric models to examine price transmission dynamics during and after the 2008 food price crisis, during the COVID-19 pandemic and the war between Russia and Ukraine. Section 5 investigates the influence of rice quality grades and consumer preferences on price transmission across the four countries studied, revealing potential price adjustments unique to each nation. Finally, Section 6 synthesises our findings and draws conclusions on the extent of market integration and price transmission observed in the Asian rice market. We conclude by offering policy implications derived from our analysis, considering the growing global population and the critical role of rice markets for producers and consumers worldwide.

2 | Price Shock Transmission in Rice Markets

According to the Law of One Price (LOP), goods that are traded in different competitive markets, without any transportation costs or trade barriers, should have exactly the same price (in common currency units) (Isard 1977). However, empirical studies have determined that the transmission of price shocks between different markets can be symmetric or asymmetric. In the case of asymmetry, the natural price adjustments differ in terms of size and timing. Asymmetric price transmission (APT) could give rise to inefficient marketing and welfare distribution. These asymmetries would cause a delay in the price adjustments at the wholesalers or producers levels (Kinnucan and Forker 1987; Fackler and Goodwin 2001; Lloyd 2017). Therefore, APT may imply food insecurity by reducing consumer benefits from lower producer prices and lower price incentives for farmers to allocate their resources to increase their production levels and income while also potentially exacerbating price fluctuations (Onumah et al. 2022; Valera et al. 2024).

In agricultural markets, imperfect price transmission in the form of asymmetries can be caused by multiple factors, including market power. Research by Meyer and Von Cramon-Taubadel (2004) and Vavra and Goodwin (2005) highlight the influence of market power on price movements across regions.

In addition, Gardner (1975) indicated that APT might be the result of public policies and government interventions. For instance, Serra and Goodwin (2003) argued that quotas implemented when downstream firms compete for market share can delay farm price transmission, leading to asymmetry. Studies by Ghoshray (2008), Fiamohe et al. (2013), Hassanzoy et al. (2016) and Sirikanchanarak et al. (2016) confirm the existence of APT in both national and international rice markets. Additional reasons for APT include menu costs that companies could face when prices change, generally correlated with inflation (Ball and Mankiw 1994), and market failures, such as structural breaks in the long-run relationship between price data characteristics and behaviour by market actors (Meyer and Cramon-Taubadel 2004).

Price transmission in the rice markets can vary depending on several factors, including market power (Deb et al. 2020) and the quality of the rice itself (Hassanzoy et al. 2016). John (2014) and Chen and Saghaian (2016) demonstrated that the global rice export market is well integrated. John (2014) found price relationships among five major rice exporting countries from Asia and the Americas, with stronger connections within Asian countries. Chen and Saghaian (2016) investigated Vietnam, Thailand and the United States (US) and confirmed market stability. They identified Thailand as a price leader that influences both prices in the US and Vietnam, while Vietnam reacted to price changes in both Thailand and the US. These results are in line with the outcomes reported by Onumah et al. (2022), who corroborated the market integration between Ghana's regional rice markets and the world market. Focusing on the influence of rice quality grades, Chizari et al. (2013) and Hassanzoy et al. (2016) found that price shocks transmit faster among low-quality rice grades, although global rice price influences are stronger and long-lasting for low-quality rice compared to high-quality. This means that higher quality rice will exhibit more stable prices during price shocks. Finally, Chulaphan et al. (2013) documented long-run relationships between high and low-quality rice (less than 20% broken rice¹ and greater than 20% broken rice, respectively) among major Asian rice exporters, implying potential spillover effects across quality grades.

This study analyses the impact of three major global shocks on rice markets: the 2008 global food price crisis, the COVID-19 pandemic and the ongoing war between Russia and Ukraine. Although these shocks originated from different economic, health related and geopolitical factors, they all significantly affected market integration and the stability of rice value chains. Analysing how these events influenced the transmission of price shocks in Asian rice markets contributes to a deeper understanding of these markets' responsiveness to external disruptions. The next section presents the data used to examine these dynamics.

3 | Data

In this study, we analyse data for the export rice prices of Thailand, Vietnam, India and Pakistan from January 2000 to June 2024. The export price data for 25% broken rice was sourced from the Food Price Monitoring and Analysis (FPMA) Tool of

the FAO (2024). We selected the price of 25% broken rice due to its relevance for food security. While higher grades such as 2% or 5% broken rice are often preferred in developing countries, price data availability for these grades is currently limited. The use of the 25% broken rice export price data is also consistent with the data used by Valera and Lee (2016). Moreover, we collected monthly oil price data from the Federal Reserve Bank of St. Louis (2023b). The oil prices utilised in this paper are crude oil prices: Brent—Europe with no seasonality adjustments in dollars per barrel. Oil prices are expected to impact the production costs and transportation between rice markets.

Due to the export ban imposed by India,^{2,3} export prices of Indian 25% broken rice were not available for the periods between 2003 and June 2004 and between 2007 and 2011, while the export restrictions imposed after August 2023 have limited the data availability for the end of the period under study. Fortunately, Indian non-basmati milled rice was still exported during those periods. Using the approach proposed by Elleby et al. (2015), we used data for non-basmati milled rice from the Live Rice Index (LRI 2022) and oil prices to impute the Indian export prices of 25% broken rice. To eliminate the influence of inflation on our analysis, we converted both monthly rice export prices and oil prices into real values by using the consumer price index of the DallasFed (2022). Consumer Price Index (CPI) data was sourced from the Federal Reserve Bank of St. Louis (2023a).

Figure 1 shows the evolution of the export prices of Indian 25% broken rice, the Indian semi-milled (or wholly milled) (IndiaSemi) price index and the oil price. Both series tend to move together during the period under analysis. Therefore, the missing values of export rice prices were estimated through a multiple imputation regression method of three-month smoothing using the prices of Indian semi-milled (or wholly milled) rice and oil. Following this approach, we imputed the data of Indian 25% broken rice and completed the series under analysis.

The series presented in Figure 1 tend to follow a similar trend, especially from 2000 to 2010 and from 2014 to 2022. The food price crisis between 2006 and 2008 created a spike in food prices worldwide, including the countries under analysis. As rice prices in different export markets tend to move with similar patterns, the factors that affect market integration in the export rice market are further analysed in this study for Thailand, Vietnam, Pakistan and India. The monthly export prices of 25% broken rice for the four countries under analysis are presented in Figure 2.

Figure 2 shows that the rice prices of the four countries studied tend to move together, which corroborates the expectation of cointegration relationships between them. The characteristics of these co-movements are further analysed in this article. Price shocks occurred in 2008 as part of the food price crisis. The differences between Thailand and Vietnam compared to India and Pakistan between 2020 and 2021 could be due to the COVID-19 pandemic-related disruptions in the chain distribution/transportation. The price variations between these two groups of countries might also be related to geographical proximities and similarities in the type of rice preferred by consumers within each group of countries.

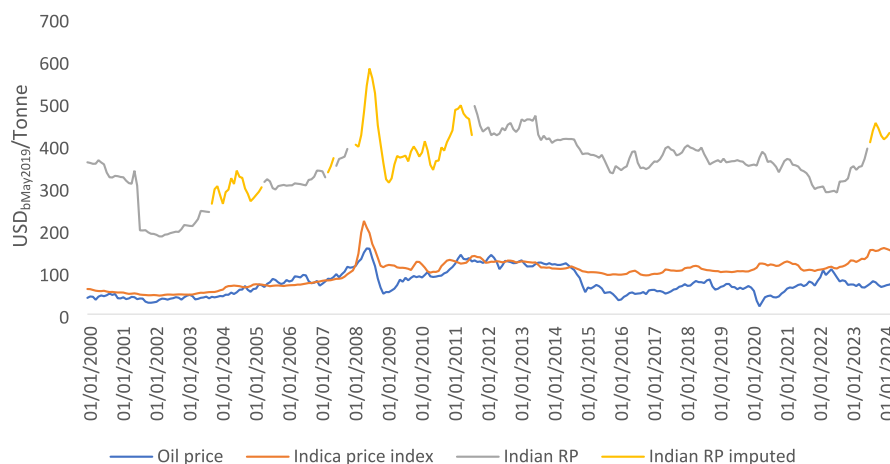


FIGURE 1 | Export prices of Indian 25% broken rice and Indian milled semi rice. Source: FAO (2024) and LRI (2022). RP, rice price.

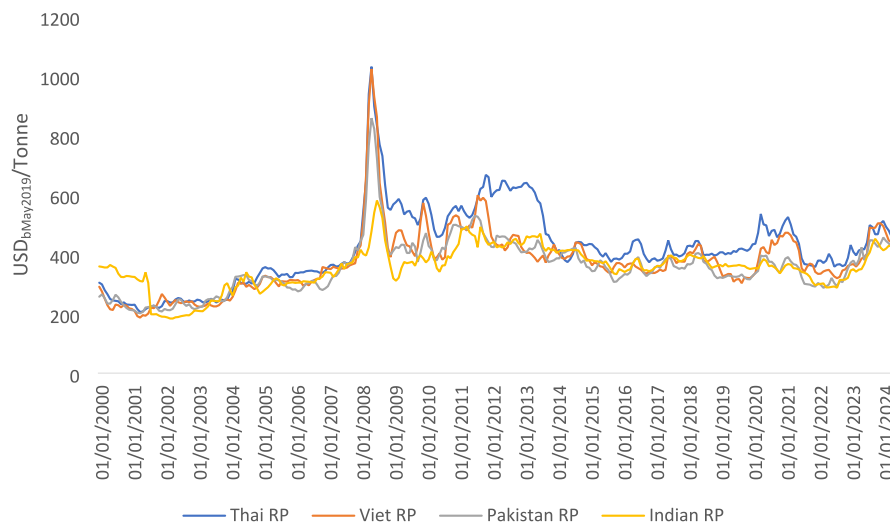


FIGURE 2 | Monthly export prices of 25% broken rice from Thailand, Vietnam, Pakistan and India. Source: FAO (2024) and LRI (2022). RP, rice price.

We analyse the relationship between monthly export data for rice and oil prices by first converting them to real values. Next, we detrend them to remove long-term trends and isolate cyclical fluctuations in the time series. We chose the Hodrick-Prescott (HP) filter for this purpose following De Jong and Sakarya (2016). This is particularly important for the rice price series, which exhibited substantial trend components, especially between 2007 and 2010 (under 5.7 years). These trends could have biased our analysis if left unaddressed, as noted by Morales et al. (2017). Detrending with a filter helps to mitigate the impact of unexpected trends, such as those observed during the 2006–2008 food price crisis, which could have resulted in less reliable estimations. Figure 3 shows the logs of monthly export price series of 25% broken rice after detrending using the HP filter. According to Maravall and Del Rio (2001), monthly price series can exhibit a cycle of reference of approximately 5.7 years and a lambda indicator of 14,400. They also highlight the widespread use of the HP filter to separate the trend and cyclical components in business cycle analysis. Consequently, they employ detrended time series data (derived using the HP filter) for their econometric modelling. This detrending process ensures that the focus is on cyclical fluctuations rather than long-term trends. Notably, these detrended variables demonstrate consistent

patterns of change throughout the analysed period, spanning from January 2000 to June 2024.

Figure 3 makes it difficult to identify a clear dominant cycle range for each country due to significant fluctuations throughout the study period. For instance, during the food price crisis between September 2008 and November 2010, India's rice price cycle was faster than that of other countries. To gain deeper insights into individual country cycles, we ran smoothed periodogram cycle analysis. This method reveals the extent to which different frequencies contribute to the price cycle variations. The results indicate that Vietnam's rice prices exhibited the highest spectrum with a 36-month periodicity and a peak spectral value of 0.149. Thailand's rice prices followed, with a dominant cycle of 41.14 months and a rice price peak spectral value of 0.1318. In contrast, India and Pakistan displayed weaker cycles, with peak spectral values of 0.0979 and 0.1227 occurring at 32 and 41.14 months, respectively. Therefore, the main rice price cycles for Vietnam, Thailand, Pakistan, and India are 36, 42, 42 and 32 months, respectively. In the specific case of Thailand and Pakistan, their rice price cycles align with the 42-month oil price cycle. These findings are worth considering when analysing the Impulse Response Functions (IRFs).

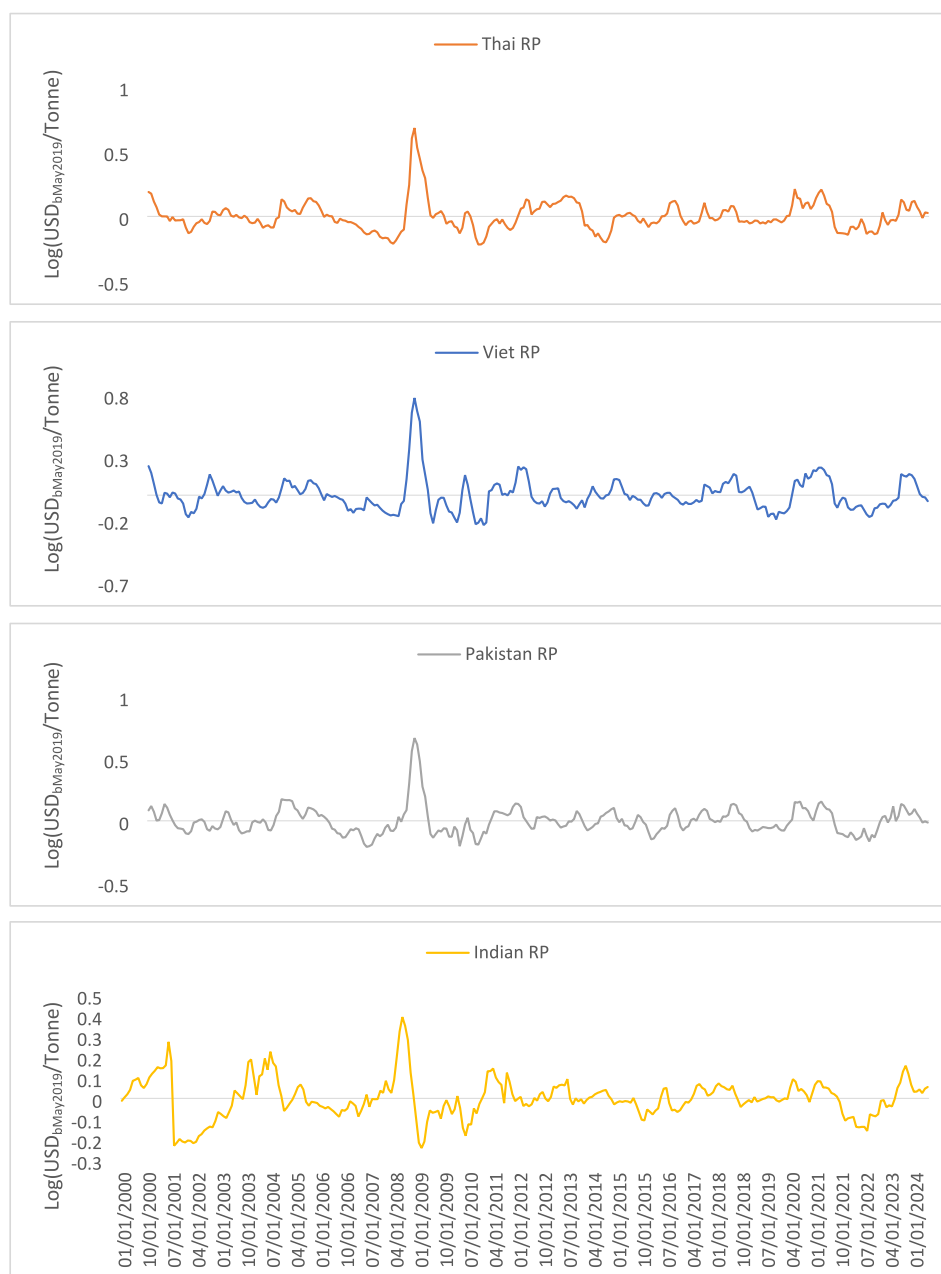


FIGURE 3 | HP detrended series of monthly export prices of 25% broken rice for the four countries analysed from 2000 to 2024. RP, rice price.

To examine the relationships between pairs of rice exporting countries, we used cross-correlation functions (CCFs) to identify the level of similarity between their time series data, following the approach of Derrick and Thomas (2004), as presented in Figure 4. The CCF analysis, shown in this figure, reveals that Thailand, Vietnam, and Pakistan exhibit the highest correlation at lag 0 (over 0.5). In contrast, Indian rice prices show a lower relationship (under 0.5) when compared to prices in other countries. Consequently, contemporaneous shocks tend to be transmitted between the rice prices of Vietnam, Thailand, and Pakistan. However, the impacts of simultaneous shocks of Indian rice prices on those of other Asian exporting countries remain unclear.

Table 1 presents the results of unit root tests conducted on the rice and oil price series. At the 1% level of significance, the HP detrended logarithm of the 25% broken real rice price series

for Vietnam, Thailand, India and Pakistan are found to be stationary at both their original level and after first differencing. Moreover, the detrended real oil price series are also found to be stationary at both levels. Consequently, both the rice and oil price series satisfy the necessary condition for employing autoregressive conditional heteroskedasticity (ARCH) and vector autoregression (VAR) models.

The dataset we described in this section presents monthly price movements across major Asian rice-exporting countries, forming the foundation for our analysis of price shocks transmission over time. Preliminary observations of the data indicate that rice prices experienced significant fluctuations in all four countries over the sample period, with prominent spikes during the 2008 global food crisis, the COVID-19 pandemic and the start of the Russia–Ukraine war. Although price movements show similar

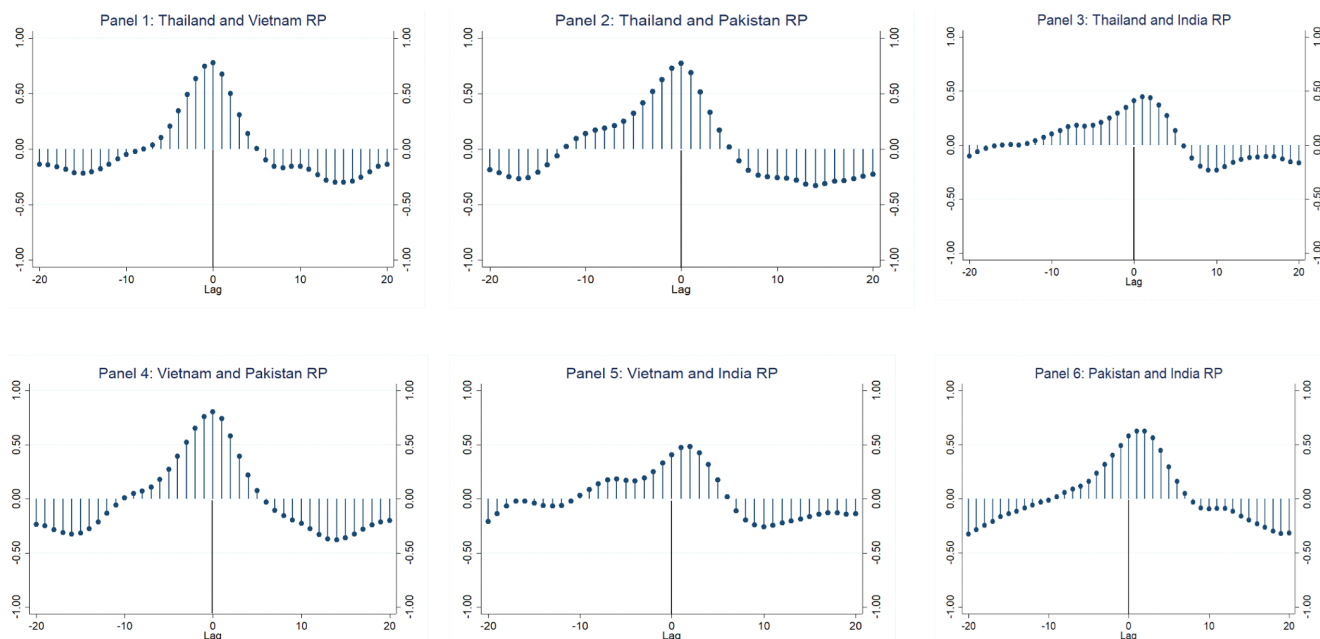


FIGURE 4 | The estimation of cross-correlation functions of monthly export prices of 25% rice broken down for the four countries analysed from 2000 to 2024 after detrending.

TABLE 1 | Results of unit root tests of HP detrended log price series of export 25% broken rice from Vietnam, Thailand, India and Thailand, and the oil price.

	Stage	ADF	Phillips-Perron	ADF-GLS
LogThaiRPrice	Level	−6.0544***	−3.9047***	−3.2508***
	First difference	−11.2542***	−10.7159***	−10.9081***
LogVietRPrice	Level	−6.9198***	−4.9154***	−3.4644***
	First difference	−10.8026***	−9.9839***	−8.0542***
LogPakRPrice	Level	−6.5001***	−4.6995***	−5.2518***
	First difference	−11.1562***	−10.2144***	−9.1199***
LogIndiaRPrice	Level	−5.6488***	−4.9461***	−5.6262***
	First difference	−13.6490***	−13.3850***	−12.7674***
LogOilPrice	Level	−6.1766***	−5.1687***	−5.9801***
	First difference	−12.6418***	−14.2444***	−3.0467***

Note: Unit root tests with intercept and no trend; *** indicates the coefficient is significant at the 1% probability level.

patterns across these markets, the timing and magnitude of price adjustments differ, suggesting varying degrees of market integration and responsiveness to external shocks. The following section outlines the empirical framework used to assess these dynamics and to identify potential effects of international and regional price shocks on rice market integration.

4 | Price Transmission Empirical Testing Framework

To analyse the extent and direction of price transmission among Asian rice markets, this study applies a structural vector autoregressive (SVAR) framework. The SVAR model

allows for the identification of structural shocks (fundamental economic disturbances) and tracks their evolving effects over time. This approach is well suited to the objectives of this study, as it enables the examination of how price shocks stemming from three distinct global events—the 2008 food price crisis, the COVID-19 pandemic and the war between Russia and Ukraine—have propagated through regional rice markets. These events serve as empirical scenarios, each with a different origin, to test the interconnectedness and responsiveness of Asian rice markets to different types of shocks. By estimating IRFs within this framework, the analysis offers valuable insights into the degree of market integration and the relative influence of international and regional shocks on rice price transmission.

Most studies on price transmission have relied on time-series data (von Cramon-Taubadel and Goodwin 2021). Several researchers employed Granger Causality (Koontz et al. 1990; Uri et al. 1993; Alexander and Wyeth 1994; Mendoza and Rosegrant 1995; John 2014), impulse response functions of vector autoregressive models (VAR) (Goodwin et al. 1999; John 2014) and cointegration techniques such as threshold error correction models (ECM) and vector ECMs (VECM) to analyse price transmissions between spatial locations (Balke and Fomby 1997; Baulch 1997; Barrett 2001; Goodwin and Piggott 2001; Chen and Saghaian 2016).

Furthermore, price volatility also needs to be considered, as it tends to transmit through price adjustments in the rice market. Natcher and Weaver (1999) found that price volatility transmits when export prices increase and decrease. Morgan (1999) estimated a volatility index for each of the price series. Setiawati and Taufikurohman (2021) recommend an autoregressive conditional heteroscedasticity (ARCH) model for estimating volatility transmission for food staples. They explained that the models could resolve the effect of time dependent heteroscedasticity in time series data. These models are appropriate to describe the time series data of the Asian rice market, due to the observed significant variations in prices during the crisis period. The ARCH model is given by this equation below:

$$P_{it}^2 = \theta_0 + \sum_{l=1}^K \theta_l (P_{i(t-l)})^2 + v_{it} \quad (1)$$

where P_{it} is the monthly price of rice exports in country i at time t , θ_0 is a constant variable, θ_l is estimated order l coefficients, K is the length of time series data, and v_{it} is the ARCH term or volatility.

Unit root tests are used to confirm the stationarity of the time series. If the data are stationary, the mean and variance do not change over time and autocorrelations and covariance between two values from the series only depend on the length of time separating the two values. In contrast, non-stationary series with non-constant means are often illustrated as not having the property of mean reversion (Hill et al. 2018). If the price series are stationary, this equation can be applied for the LOP test with reliable results (Enders 2014; Greene 2018).

VAR is a common model for testing the interdependency among time series variables. This model is useful for this paper because it allows testing the price transmission dynamics among rice prices, using endogenous and, in some cases, exogenous variables (Pesaran and Smith 1998). Furthermore, the VAR is applied in this article to explore the size and dynamics of price transmission (Morales et al. 2017). In addition to providing standard estimation outputs, VAR models hold special capabilities for uncovering interactions between variables. These capabilities include features like the impulse response function (IRF), which represents the reaction of a variable to an input signal (Uhlig 2005). Finally, the stochastic trend is often explored in price variables, resulting in non-stationarity during sample time, and in a VAR model, we can utilise the non-stationarity of variables to study the main driving trends

to the variables (Kaspersen and Føyn 2010). Our model incorporates dummy variables to capture the independent effects of the food price crisis in 2008, the COVID-19 pandemic and the war between Russia and Ukraine on the transmission of rice price shocks. Therefore, the dummy variable for the food price crisis takes a value of 1 from January to December 2008 and 0 otherwise. Similarly, the dummy variable for the COVID-19 pandemic takes a value of 1 for the period from January 2020 to June 2021, and 0 otherwise. Finally, the dummy variable for war represents the period from February 2022 to June 2024, following the start of the war between Russia and Ukraine, which remains ongoing. This approach allows us to isolate and estimate the specific impact of each event on rice price transmission within the model.

The VAR model with i stationary price series is shown below:

$$\ln P_{it} = \omega_i + \sum_{i=1}^4 \sum_{s=1}^K \beta_{is} \ln P_{i(t-s)} + \sum_{i=1}^4 \alpha_i \text{vol}(\ln P_{it}) + \tau_i \ln \text{Poil}_t + \varphi_{i1} D_{FC} + \varphi_{i2} D_{COVID} + \varphi_{i3} D_{WAR} + e_{it} \quad (2)$$

where P_{it} is $j \times 1$ price of rice grade in country i at time t ; ω_i is a $j \times 1$ constant of country i ; β_{is} are $j \times 1$ parameters; Poil_t is oil price at time t ; K is the number of lags; D_{FC} , D_{COVID} and D_{WAR} are dummy variables for the food price crisis, the COVID-19 pandemic and the conflict between Russia and Ukraine; and e_{it} is the error ($E(e_{it})=0$).

Canova (2007) highlighted the reliability of structural VAR (SVAR) for analysing structural economic shocks in time series data. Gottschalk (2001) considers it the ‘nearest precise’ approach in empirical shock analysis. Pfaff (2008) emphasises the model’s suitability for investigating contemporaneous effects between variables based on explicit modelling. Morales et al. (2017) further contribute by demonstrating SVAR’s ability to explain the time-varying influence of shocks on price in separate markets through VAR application. They also emphasised the importance of imposing restrictions on the error terms to accurately capture structural shocks. Essentially, SVAR models can be implemented within the VAR framework, but with additional identifying restrictions.

Applying the AB model of Amisano and Giannini (2012), we impose A and B matrices in Equation (2) above and the formulation to add restrictions on the SVAR model, which is presented below:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \text{NA} & 1 & 0 & 0 \\ \text{NA} & \text{NA} & 1 & 0 \\ \text{NA} & \text{NA} & \text{NA} & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} \text{NA} & 0 & 0 & 0 \\ 0 & \text{NA} & 0 & 0 \\ 0 & 0 & \text{NA} & 0 \\ 0 & 0 & 0 & \text{NA} \end{bmatrix}$$

$$Ae_t = Bu_t \quad (3)$$

where the structural shocks u_t are normally distributed; e_t is the vector of errors; A is a restriction identification matrix and B is an diagonal matrix; “NA” defines that there is an unknown value of the coefficient in the matrix that has to be estimated.

TABLE 2 | VAR model lag selection.

Lags	LogL	LR	FPE	AIC	SC	HQ
0	1356.77005	NA	1.00e−09	−9.36716	−8.90223	−9.18072
1	2027.64738	1279.901	9.66e−12	−14.0116	−13.3401	−13.7423
2	2113.06984	160.5458	5.91e−12 ^a	−14.504 ^a	−13.625 ^a	−14.151 ^a
3	2126.26979	24.43395	6.03e−12	−14.4841	−13.3993	−14.0491
4	2142.44984	29.4913	6.03e−12	−14.4854	−13.194	−13.9675
5	2155.77524	23.91012	6.15e−12	−14.4664	−12.9684	−13.8657
6	2170.8964	26.70341	6.19e−12	−14.4602	−12.7555	−13.7766
7	2185.2155	24.8807	6.28e−12	−14.4483	−12.5369	−13.6818
8	2200.0014	25.27239	6.35e−12	−14.4397	−12.3217	−13.5903
9	2204.6256	7.772605	6.90e−12	−14.3590	−12.0344	−13.4268
10	2222.2037	29.04749	6.85e−12	−14.3702	−11.8389	−13.3551
11	2242.8795	33.5798 ^a	6.65e−12	−14.4034	−11.6655	−13.3054
12	2250.8878	12.77908	7.07e−12	−14.3467	−11.4022	−13.1659

Abbreviations: AIC, Akaike information criterion; FPE, final prediction error; HQ, Hannan-Quinn information criterion; LogL, log likelihood; LR, sequential modified LR test statistic (each test at 5% level); SC, Schwarz information criterion.

^aIndicates lag order selected by the criterion, 1e−09 and 1e−12 equals 1×10^{-9} and 1×10^{-12} .

TABLE 3 | VAR Granger causality test for the first difference of HP detrended log rice export prices for the four countries analysed.

Chi-squared values of	LogThai RPrice	LogViet RPrice	LogPak RPrice	LogIndia RPrice	All
LogThaiRPrice responds to	—	8.0864**	2.1248	3.5584	14.1646**
LogVietRPrice responds to	14.744***	—	15.775***	1.2780	24.903***
LogPakRPrice responds to	9.07261**	9.4395***	—	3.9654	35.518***
LogIndiaRPrice responds to	0.0550	2.9283	4.9817*	—	17.537***

Note: ***, ** and * display that the coefficient is significant at the 1%, 5% and 10% probability levels, respectively.

In the last estimation using EViews 14, we rely on the impulse response to a structural shock followed by the interaction between VAR and SVAR shock of $Ae_t = Bu_t; (\epsilon_{it})$ and the standard deviation of shocks (Ouliaris et al. 2016); $E(u_t u_t') = I$; and B is a diagonal matrix.

The modelling approach presented above provides the empirical foundation for assessing price transmission dynamics across Asian rice markets. The following section presents the empirical results, focusing on how markets respond to international and regional shocks.

5 | Empirical Results

This section presents the main results of our price transmission analysis across Asian rice markets. First, we establish the optimal lag structure for the VAR model. Then, we examine how rice prices dynamically respond to global shocks. Finally, we present causality tests and impulse response functions that shed light on the extent of market integration and the patterns of price adjustment among the countries studied. The first step to run

the VAR model is to identify the number of lags that should be included. In this case, the maximum number of lags is 12, due to the nature of time series data (monthly). Table 2 presents different criteria to select the number of lags.

Table 2 indicates that the optimal number of lags for this model is two, based on the FPE, AIC, SC and HQ criteria. In addition, we employed the VAR Granger causality test to determine the direction of the price transmission in each country. Table 3 presents the relationships between the export prices of 25% broken rice for Vietnam, India and Pakistan.

The Granger causality test assesses the relationships between Vietnam, Thailand, India and Pakistan export rice prices following Equation (2). These results also help to determine bi-directional price transmission in the Asian rice market in the short run that are presented in the reduced form VAR model in Table 4. The results of Table 3 show that price shock transmission exists among the export rice prices of the four countries under analysis. Furthermore, the chi-squared values between Vietnam and Pakistan export rice prices, and between Vietnam and Thailand export rice prices are significant at the 1% and 5%

level of significance in both directions. This result demonstrates that rice price shocks are transmitted between Asian countries exporting low-quality rice. In contrast, the chi-squared values of Indian export rice prices are not significant with respect to the export rice prices of other countries, which corroborates the lack of relationships between Indian export rice prices and rice prices from other countries analysed and presented in Figure 4. This outcome could be due to the policies implemented by the Indian government to insulate the rice prices in the domestic market. The relationship between Pakistan and Indian export rice prices likely stems from their geographical proximity and the strong consumer preference for basmati rice, a food staple in both countries. The results of the reduced form VAR model are presented in Table 4.

The reduced form VAR model indicates that the dummy variable of the food price crisis that captures the period from January to December 2008 was significant for the export rice prices of Thailand and Pakistan. Meanwhile, the COVID-19 dummy variable that covers the period from January 2020 to June 2021 was significant for the Vietnam and Pakistan export rice prices. This outcome may be attributed to disruptions in domestic agro-food chains caused by financial services and operational challenges during the food price crisis and the COVID-19 pandemic.

Conversely, the oil price and the dummy variable 'war' are not significant variables influencing price shock transmission. The first result suggests that oil prices do not affect price movements in the Asian rice market. The result of the dummy variable 'war' in the last period could be explained by less direct impacts of the war on the performance of domestic agro-food chains, in contrast to the food price crisis and the COVID-19 periods. Therefore, these results demonstrate that during the periods of crises that directly affect the operations of domestic agro-food chains, there have been impacts on price transmission between the four rice exporting countries.

With respect to price volatility, Thailand is affected by its own export rice price volatility, while Pakistan export rice prices are significantly affected by fluctuations in Vietnam export rice prices at the 5% level of significance, but not by variations in rice prices originating in other countries. This aligns with the results of the CCF reported in Figure 4 which shows India's rice prices with the weakest link to others. Interestingly, Pakistan rice prices respond to shocks in Vietnam and Thailand rice prices within the first period ($p < 0.05$), whereas Vietnam and Thailand rice prices do not react to Pakistan's prices in the first three periods. Finally, the reduced form VAR model was estimated for the sub-period January 2008–June 2024 to control for a potential structural break related

TABLE 4 | Reduced form VAR model of HP detrended log rice export prices of Vietnam, Thailand, India and Pakistan (2000–2024).

	LogThaiRPrice	LogVietRPrice	LogPakRPrice	LogIndiaRPrice
LogThaiRPrice ($t-1$)	1.1169***	0.1758*	0.2099***	0.0045
LogThaiRPrice ($t-2$)	−0.3248***	−0.3003***	−0.2333***	−0.0128
LogVietRPrice ($t-1$)	0.1733**	1.0908***	0.1752***	0.1058*
LogVietRPrice ($t-2$)	−0.1343*	−0.3733***	−0.1633***	−0.0932
LogPakRPrice ($t-1$)	0.0927	0.0981	1.0321***	0.0494
LogPakRPrice ($t-2$)	−0.1004	0.1358	−0.2744***	0.0664
LogIndiaRPrice ($t-1$)	−0.1039	−0.0192	0.0046	1.0660***
LogIndiaRPrice ($t-2$)	0.1180*	−0.0264	0.0594	−0.2995***
Constant	−0.0055	−0.0085*	−0.0061	−0.0032
LogThaiRPriceVol	−3.7847**	−1.5554	−1.5010	−0.5630
LogVietRPriceVol	0.3557	−1.0767	1.0907	−0.0324
LogPakRPriceVol	4.0571	5.4680**	1.3622	1.8990
LogIndiaRPriceVol	−0.4306	−0.3250	−0.2811	0.2727
LogOilPrice	−0.0122	−0.0051	0.0131	0.0197
Dummy food price crisis (2008)	0.0496***	0.0258	0.0362**	−0.0205
Dummy COVID-19 (January 2020–June 2021)	0.0173	0.0336**	0.0240**	0.0055
Dummy war (February 2022–June 2024)	0.0026	−0.0052	0.0003	−0.0012
Within R^2	0.8822	0.8732	0.8804	0.8184
Observations	292 after adjustments			
AIC	−14.556			
SIC	−13.7002			

Note: ***, ** and * display that the coefficient is significant at the 1%, 5% and 10% probability levels, respectively. Abbreviations: AIC, Akaike information criterion; SIC, Schwarz information criterion.

to the food price crisis, but the result demonstrates that there is no evidence of a structural break.

To further analyse how export rice price shocks are transmitted between exporting countries, we use generalised impulse response functions (GIRFs) based on SVAR modelling. This approach captures potential contemporaneous effects among exporting countries. The contemporaneous effects can be tested using the Cholesky decomposition of the causal defined ordering of shock propagation from countries mostly consuming jasmine rice (Thailand and Vietnam) to countries that mostly consume basmati rice (Pakistan and India). The results of the Granger causality test presented in Table 3 highlight an influence of Thailand and Vietnam export rice prices on Pakistan rice prices, while only the prices of Pakistan have an influence on Thailand and Vietnam export rice prices. A robust test using a change in the ordering of the variables is further implemented to explore the opposite case. Considering the well-defined variable ordering in the SVAR representing the causal structure of the shock transmission, the Cholesky decomposition is a useful approach for this study. We run SVAR models with different order groups of matrix A following the Cholesky decomposition approach to test the reliability of the results obtained. This is required due to the identification scheme used by this method, which creates a recursive and contemporaneous ordering of variables. This can make the results dependent on the order used during the estimation (Uhlig 2005). The Cholesky decomposition only allows estimating the contemporaneous response to shocks in one direction, using a recursive restriction on the

direction of the shocks applied by the Cholesky identification with a lower triangular variance-covariance matrix with restrictions applied to the residuals. However, the selection of restrictions would impact the estimation of the model. Hence, Sims (1986) and Bernanke (1986) suggested adjusting the ways of identifying restrictions. It is the approach we followed in this research. Therefore, in these estimations, two group orders of rice price series are imposed as restrictions in the IRFs to ensure a comprehensive representation of all shocks. In the first group order, Thailand is at the beginning, followed by Vietnam, Pakistan and India. The second group order was implemented to test for robustness, with Pakistan rice prices included first in the model, followed by India, Thailand, and Vietnam. Following Equation (3), we estimate the IRFs under SVAR and present the results in Figure 5. Considering that the maximum range of cycle contribution of rice prices is 42 months for the four countries analysed, we apply an optimisation trace period of 42 for structural VAR and 42 months for IRF.

Figure 5 shows that rice prices react more to price shocks in their own country than to shocks on rice prices in other countries, as shown in Panels 1, 6, 11 and 16 of Figure 5. This could be due to these exporting countries ensuring domestic rice supply and price stability before exporting rice (John 2013; Valera et al. 2024). In addition, there are signs indicating that Asian rice markets have similar reactions. Initially, rice prices respond swiftly to shocks, with fluctuations diminishing later. During the first period, their own shocks trigger an initial positive response, followed by a sharp decline. The responses to shocks last under 10 months,

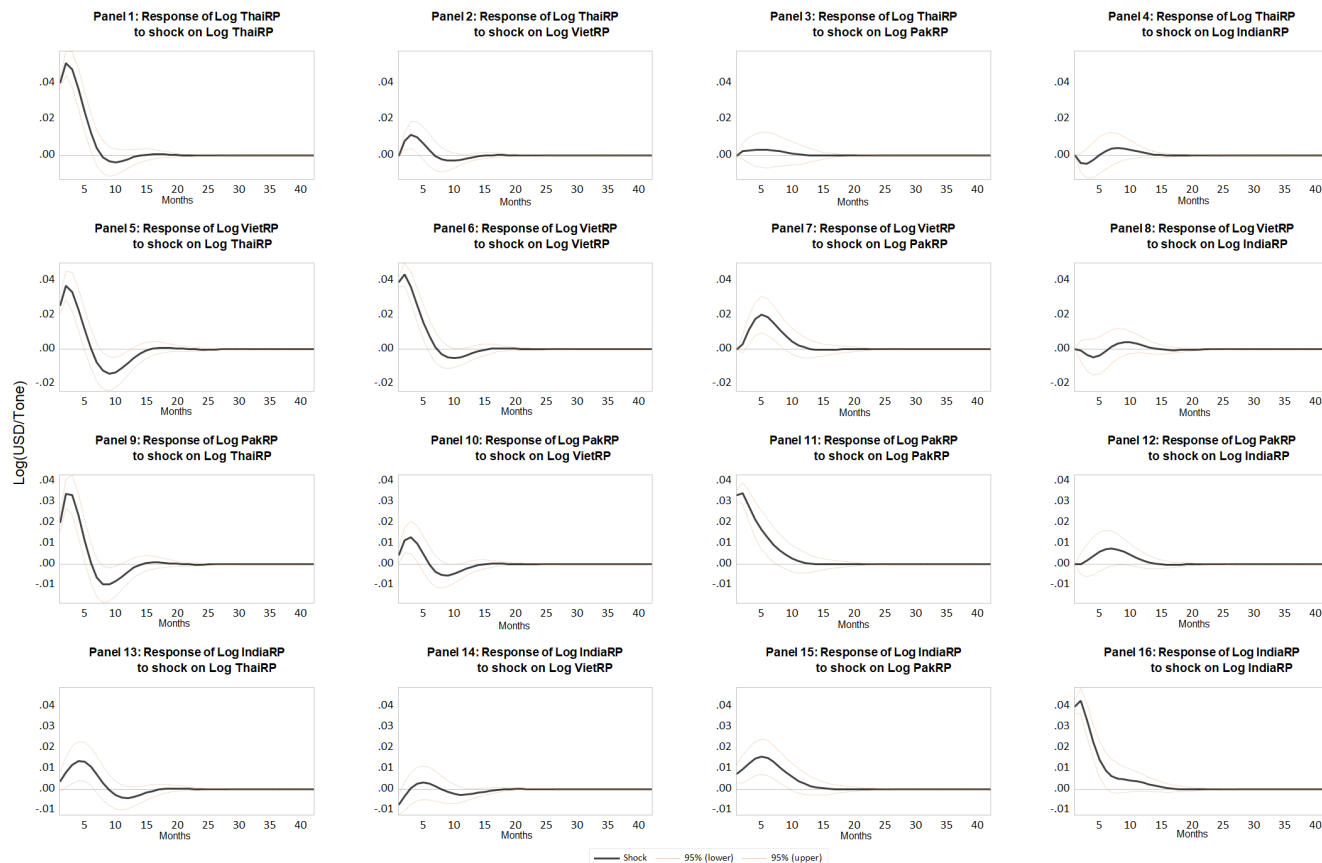


FIGURE 5 | IRFs of HP detrended log price series of export 25% broken rice of Thailand, Vietnam, Pakistan and India under SVAR.

which is less than the over 10 months observed in India and Pakistan. Moreover, India and Pakistan rice prices significantly fluctuate around zero after 10 months, unlike the stable rice price decline observed in Thailand and Vietnam. Interestingly, Vietnam and Thailand exhibit similar responses in direction and magnitude to their own shocks throughout the period.

Analysing the transmission of price shocks between countries, we found that Thailand and Pakistan rice prices tend to react similarly to shocks from Vietnam rice prices, as shown in Panels 2 and 10 of Figure 5. Moreover, we observe high variations in Pakistan rice prices resulting from shocks in Thailand, as presented in Panel 9 of Figure 5. In the case of Vietnam and Thailand, rice prices tend to respond significantly more in the first period and shocks are completely absorbed after 15 months, as depicted in Panels 2 and 5 of Figure 5. In contrast, Panels 12 and 15 of Figure 5 illustrate that Indian and Pakistan rice prices show a quick fluctuation in the first 15 months, suggesting high sensitivity to shocks affecting each other's markets. However, it is worth noting that neither Vietnam nor Thailand consume or export basmati rice. Despite this difference, Pakistan export rice prices exhibit a similar response to shocks in export rice prices from Thailand, as Vietnamese rice prices do (Panels 5 and 9 of Figure 5). Similarly, Thailand and Pakistan rice prices respond to price shocks in Vietnam, as illustrated in Panels 2 and 10 of Figure 5. However, India export rice prices exhibit a distinct pattern. Indian rice prices remain largely unaffected by price fluctuations in the other three countries we studied. This divergence likely stems from India's specific price and trade policies, which influence how market shocks are transmitted within its rice market. Therefore, these findings suggest that low rice quality has no impact on price transmission within the Asian rice market. As shown in Table 3, the VAR Granger causality test results indicate that only Indian rice prices respond to shocks in Pakistan rice prices. Conversely, Pakistan, Vietnam and Thailand rice prices seem to respond similarly to shocks in other countries. This unique behaviour of Indian rice prices can be attributed to their trading policies during crises, as seen above. This outcome aligns with the CCF findings presented in Figure 4. For robustness, we conducted further estimations. First, we added oil prices as an endogenous variable. Second, we estimated the VAR model using Thailand, Vietnam and Pakistan rice prices. Across all these variations, the IRFs results obtained in Figure 5 remained consistent for all analysed countries, further supporting our initial findings.

The results presented in this section reveal that rice prices respond differently to shocks across Asian countries, reflecting variations in trade relationships, market structures and national policies. These dynamics highlight the importance of understanding how market integration and national policy decisions influence the stability of rice markets across the region. The next section draws on these findings to explore their practical policy implications and suggest directions for future research.

6 | Conclusions

Rising food insecurity and malnutrition pose a major threat to the well-being of poor and vulnerable populations, and it affects the overall societal stability. Recent economic shocks, trade restrictions

and diverse price policies in both exporting and importing nations have fuelled these price fluctuations (Ferguson and Ubilava 2022; FSIN and the Global Network Against Food Crises 2023). These volatile market conditions raise critical questions about the efficiency of staple food markets: Can they effectively transmit price signals and incentives across borders, or do price shocks remain isolated, amplifying food insecurity in some regions while leaving others unaffected? This research tackles this vital issue by examining rice price transmission among selected Asian exporting countries, focusing on the effects of crises on low-quality rice grades, including the impact of the food price crisis, the COVID-19 pandemic and the war between Russia and Ukraine.

This article studies the price adjustment of low-quality rice based on the 25% broken rice, which is primarily imported by developing countries. This study demonstrates that the rice markets of Thailand, Vietnam and Pakistan are integrated, with shocks on rice prices in one country being transmitted to others. In addition, Indian rice prices show a lower correlation with price variations in other countries, likely due to the Indian government's policies aimed at insulating domestic rice prices. Our results are consistent with the findings reported by Ghoshray (2008) regarding price shock transmission between Vietnam and Thailand rice export prices. Hence, government price or trade interventions targeting specific rice quality grades can influence domestic prices, potentially leading to negative impacts for food security (John 2013; Onumah et al. 2022; Valera et al. 2023).

Indian rice prices have remained largely unaffected by fluctuations in other major Asian rice exporting countries. Our analysis reveals that government price and trade policies significantly influence the Indian rice market, shielding it from external price shocks. While these policies may initially seem beneficial, their impact can be counterproductive. By distorting market signals and reducing transparency, they can exacerbate price fluctuations during periods of instability, threatening regional food security.

Our study demonstrates that crises can disrupt rice price transmission when they directly impact operations of domestic agro-food chains. This is evident in the cases of the food price crisis and the COVID-19 pandemic, which had clear impacts. However, the conflict between Russia and Ukraine appears to have little impact on the price transmission within domestic rice chains. Therefore, we recommend that nations prioritise targeted policies to incentivise rice production and promote market-driven mechanisms to ensure more resilient and stable rice markets in future crises.

The key limitation of this study is its focus on low-quality rice, which was analysed for only four Asian rice exporting countries. Future research could explore several avenues to address this limitation. First, comparing high-quality (5% broken rice) and low-quality (25% broken rice) would contrast price interactions between different grades. Second, expanding the analysis to include more countries involved in the global rice market for different rice qualities would provide a broader picture of price transmission for various rice qualities. Finally, exploring these price interactions in different geographical regions apart from those covered in this study would offer a more comprehensive understanding of global rice market dynamics. These

explorations would contribute to a deeper understanding of complex rice market dynamics and the impact of price fluctuations on stakeholders in diverse locations, especially rice producers and consumers worldwide, who disproportionately depend on rice for their food security, nutrition and livelihoods.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

- ¹ Basmati and jasmine rice varieties are usually categorised as high-quality rice.
- ² In October 2008, India introduced a total restriction on exports of basmati rice. This rice trade policy was partially lifted after 3 years.
- ³ India has again restricted the export of non-basmati white rice from 20 July 2023.

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